8. Collection Bulk Operations — Quick Notes

Bulk Operations

Purpose

Perform operations on:
multiple elements / collections

Available in: Collection Interface

Works for:

- ArrayList
- LinkedList
- HashSet
- TreeSet
- ArrayDeque

containsAll()

Purpose

Checks whether all items of another collection exist.

Example:

```
List<Integer> l1 = List.of(10,20,30);
List<Integer> l2 = List.of(10,30);

System.out.println(l1.containsAll(l2));
```

Output: true

Important Point

Duplicates don't matter.

Example:

```
l2 = List.of(10,30,30);
```

Still: true

Because: only presence checked

addAll()

Purpose

Adds all elements from another collection.

Example:

```
List<Integer> l1 = new ArrayList<>();

l1.add(10);
l1.add(20);

List<Integer> l2 = List.of(30,40);
l1.addAll(l2);
System.out.println(l1);
```
Output: [10, 20, 30, 40]

Return Value of addAll()

Returns:

- `true` → collection modified
- `false` → no changes

Example:

```
Set<Integer> s = new HashSet<>();
s.add(10);
s.addAll(List.of(10));
```

Returns: false

Because: Set already contains 10

addAll() Generic Syntax

```
addAll(Collection<? extends E> c)
```

Meaning:

- Can add descendants of E

removeAll()
Purpose

Removes matching elements.

Example:

```
List<Integer> l1 = new ArrayList<>();

l1.add(10);
l1.add(20);
l1.add(30);

List<Integer> l2 = List.of(10,20);
l1.removeAll(l2);
```

System.out.println(l1);

Output:[30]

Important Point

removeAll(Collection<?> c)

Uses: equals()

So: any datatype allowed

retainAll()

Purpose

Keeps only matching elements.

Example:

```
List<Integer> l1 = new ArrayList<>();

l1.add(10);
l1.add(20);
l1.add(30);

List<Integer> l2 = List.of(10,20);
l1.retainAll(l2);
System.out.println(l1);
```

Output:

[10, 20]

removeIf()

Purpose

Remove elements using condition.

Example:

```
List<Integer> l = new ArrayList<>();

l.add(10);
l.add(15);
l.add(30);
l.add(19);

l.removeIf(n -> n % 2 == 0);
System.out.println(l);
```

Output: [15, 19]

Predicate
Functional Interface.
Contains: test()
Returns: boolean

Lambda Expression
n -> n % 2 == 0
Meaning: remove even numbers

Lambda Syntax
parameter -> logic
Example: n -> n > 15
Removes: all numbers > 15

Important Points

Method	Purpose
containsAll()	Checks presence
addAll()	Adds all items
removeAll()	Removes matching items
retainAll()	Keeps matching items
removeIf()	Removes by condition

Quick Notes

Topic	Point
Bulk Operations	Present in Collection interface
removeIf()	Uses Predicate
Lambda	Added in Java 8
Predicate	Functional Interface
addAll()	Uses `<? extends E>`
removeAll()/retainAll()	Use equals()